Macroscopic rheology of complex fluids: from viscoelasticity to elasto-viscoplasticity

A pedagogical tutorial

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Flows of Complex Fluids: from theory to applications



Roadmap

1. **Macroscopic behaviours:** shear thinning, elasticity, yield stress
2. **Linear viscoelasticity:** Maxwell, Kelvin–Voigt, then the step to the Oldroyd-B model
3. **Viscoplasticity:** Bingham, Herschel–Bulkley — and where they fail
4. **Structural models:** Putz & Burghelea; de Souza Mendes & Thompson
5. **A particle view:** yielding with no structure parameter
6. **The Saramito model** tested on an unsteady channel flow

The question that runs through this tutorial

What is the *simplest* constitutive equation that still captures the behaviour we measure?

Slides tagged **our work** show data from our own papers. Curves labelled "model" are solutions computed for this tutorial, with illustrative parameters.

LTeN

1. Macroscopic rheological behaviours

What does a complex fluid do that a Newtonian fluid does not?

The language: stress, rate of strain, material functions

Momentum balance, for every model in this tutorial:

$$\rho\left(\partial_t \mathbf{u} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \nabla\cdot\boldsymbol{\tau}, \qquad \nabla\cdot\mathbf{u} = 0$$

Rheology is the closure: $\boldsymbol{\tau} = ?$

Newtonian reference:

$$\boldsymbol{\tau} = 2\eta\mathbf{D}, \qquad \mathbf{D} = \frac{1}{2}\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right)$$

Steady simple shear, $u_x = \dot{\gamma}\, y$

Three functions of $\dot{\gamma}$ fully describe the response:

$$\eta(\dot{\gamma}) = \tau_{xy}/\dot{\gamma}$$
$$N_1(\dot{\gamma}) = \tau_{xx} - \tau_{yy}$$
$$N_2(\dot{\gamma}) = \tau_{yy} - \tau_{zz}$$

Newtonian fluid: $\eta = \text{const.}$, $N_1 = N_2 = 0$.

Three dimensionless numbers, four useful tests

Dimensionless numbers

$\mathrm{De} = \lambda / t_{\mathrm{flow}}$	memory vs. forcing time
$\mathrm{Wi} = \lambda \dot{\gamma}$	elastic vs. viscous stress
$\mathrm{Bn} = \tau_y / (K \dot{\gamma}^{\,n})$	yield vs. viscous stress

Viscoplastic flows need care with the Reynolds number too: a consistent scale is $\mathrm{Re} = \rho U^2 / (\tau_y + K \dot{\gamma}^{\,n})$ (Thompson & Soares, 2016).

Test	**What it reveals**
Steady flow curve	$\eta(\dot{\gamma})$, τ_y
Up/down stress ramp, t_0 per step	hysteresis, time dependence
Creep, start-up of shear	transients, overshoot
Oscillations (SAOS, LAOS)	G', G'', nonlinearity

Behaviour 1: shear thinning

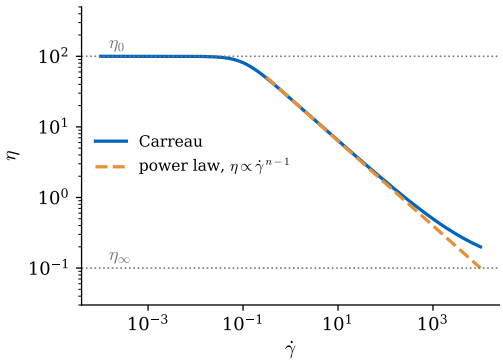
The viscosity decreases with the rate of shear, because the flow aligns, stretches or disentangles the microstructure.

Power law: $\eta = K\dot{\gamma}^{n-1}, \quad n < 1$

Carreau: $\eta = \eta_\infty + \dfrac{\eta_0 - \eta_\infty}{[1 + (\lambda\dot{\gamma})^2]^{(1-n)/2}}$

Flow consequence: a blunter channel profile,

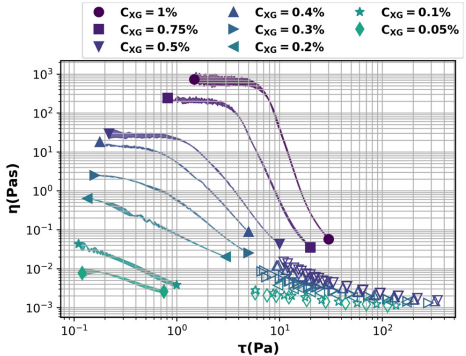
$$u/u_{\max} = 1 - |y/h|^{1+1/n}$$



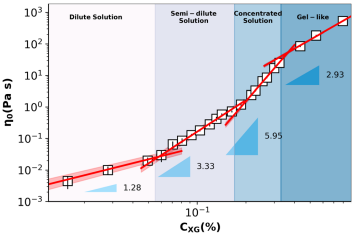
Model curve. The power law diverges as $\dot{\gamma} \to 0$: it has no Newtonian plateau.

Shear thinning in xanthan solutions

our work



Steady-state viscosity vs. stress, rotational and microfluidic rheometry. Missi *et al.*, Soft Matter 20 (2024), Fig. 3.



Zero-shear viscosity: four concentration regimes. Same paper, Fig. 10.

The same polymer goes *from shear thinning to elasto-viscoplastic* as concentration rises: a sharper and sharper drop of η at a critical stress.

Behaviour 2: elasticity (the fluid remembers)

Stress depends on the *history* of deformation over a time λ. Typical signatures:

- Stress relaxation after a flow stops
- Normal stresses, $N_1 \neq 0$: rod climbing, hoop stress along curved streamlines
- Stress overshoot at start-up of shear
- Elastic recoil when the load is removed
- Strain hardening in extension
- Purely elastic instabilities at vanishing inertia

Rule of thumb

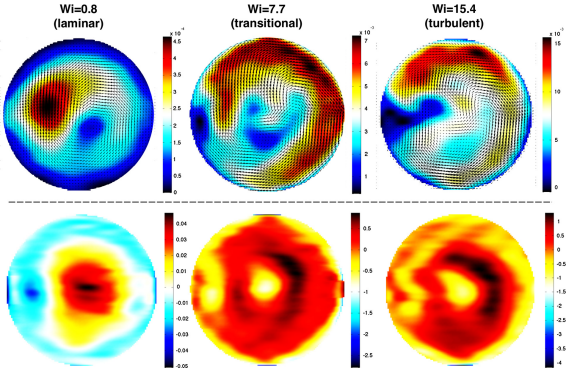
$\mathrm{De} \ll 1$: liquid-like
$\mathrm{De} \gg 1$: solid-like

The same material can be either, depending on how quickly it is forced.

Do not confuse elasticity (a stress that relaxes) with thixotropy (a structure that evolves). Both produce memory.

Elasticity at vanishing inertia: elastic turbulence

our work



Polyacrylamide in aqueous sucrose, von Kármán flow: time-averaged flow (top) and vorticity (bottom), at $\mathrm{Wi} = 0.8$ (laminar), 7.7 (transitional) and 15.4 (turbulent). Traore *et al.*, JNNFM 223 (2015), Fig. 5.

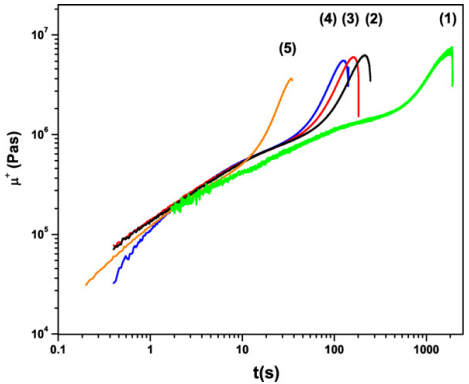
The flow is driven at a Reynolds number where the pure solvent shows no inertial instability, yet:

- above Wi_c, power injection and fluctuations rise
- heat transfer rises locally up to about $4\times$ the conductive state

Elastic stresses alone destabilise the flow.

Elasticity in extension: strain hardening of a polymer melt

our work

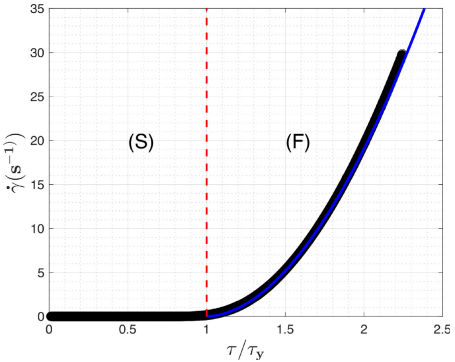


Transient elongational viscosity of LDPE at several constant strain rates.
Burghelea, Starý & Münstedt, JNNFM 166 (2011), Fig. 2.

- At short times all rates share one curve: the linear viscoelastic envelope (Trouton: $\mu^+ \to 3\eta^+$)
- At larger strains, viscosity rises well above that envelope: strain hardening
- The shape then depends on rate, and the sample eventually necks and ruptures

Hold on to this picture: Oldroyd-B also predicts growth in extension.

Behaviour 3: yield stress



Carbopol gel, slow ramp, with a Herschel–Bulkley fit. Younes *et al.*, JNNFM 281 (2020), Fig. 1.

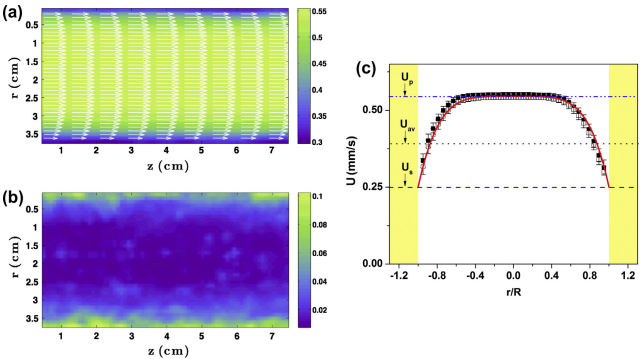
- Below a critical stress τ_y: deforms but does not flow
- Above τ_y: flows, usually shear thinning
- Everyday examples: gels, pastes, emulsions, cement, mud, mayonnaise

The historical debate

"The yield stress myth?" (Barnes & Walters, 1985): is τ_y only an instrument limit? Modern rheometers support a true yielding behaviour, *but* measured values still depend on the protocol.

Yield stress in a flow: plug in a pipe

our work

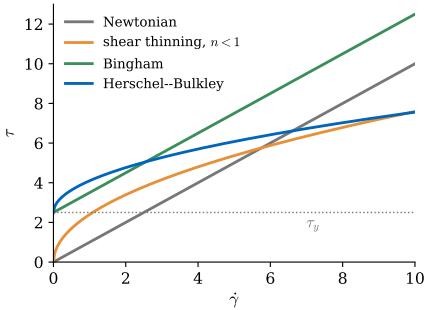


Carbopol in a pipe, particle image velocimetry. (a) Mean velocity field; (b) reduced fluctuations; (c) velocity profile with its fit. Poumaere *et al.*, JNNFM 205 (2014), Fig. 4.

- Flat central unyielded plug, where $|\tau| < \tau_y$
- Sheared annulus near the wall
- Finite velocity at the wall: these gels slip on smooth surfaces

Slip affects both the measurement of τ_y and how it should be modelled.

Interlude: real materials show all three at once



Model curves. Yield stress and shear thinning combine naturally in a flow curve.

Material	Thin.	Elast.	τ_y
Dilute PAAm	weak	✓	–
LDPE melt	✓	✓	–
Xanthan, dilute	weak	weak	–
Xanthan, high conc.	✓	✓	✓
Carbopol gel	✓	✓	✓

The model you need depends on the material *and* the question.

2. Introduction to linear viscoelasticity

Maxwell and Kelvin–Voigt, then the step to a frame-indifferent nonlinear model: Oldroyd-B

Linear viscoelasticity: the superposition principle

For small deformations, responses add up (Boltzmann):

$$\tau(t) = \int_{-\infty}^{t} G(t - t')\,\dot{\gamma}(t')\,\mathrm{d}t'$$

The relaxation modulus $G(t)$ contains the full linear response.

- Hookean solid: $G(t) = G$
- Newtonian fluid: $G(t) = 2\eta\,\delta(t)$ (with $\int_0^\infty \delta = \frac{1}{2}$)
- Viscoelastic material: $G(t)$ decays over time

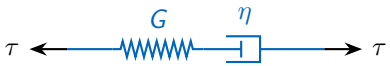
Validity

Small *strain* or small De: microstructure stays close to equilibrium.

$$\eta_0 = \int_0^\infty G(s)\,\mathrm{d}s$$

Zero-shear viscosity: the area under $G(t)$.

The Maxwell model: spring and dashpot in series



Same stress in both elements, strain rates add:

$$\dot{\gamma} = \frac{\dot{\tau}}{G} + \frac{\tau}{\eta}$$

With $\lambda = \eta/G$:

$$\lambda\,\dot{\tau} + \tau = \eta\,\dot{\gamma}$$

Solving it for τ:

$$\tau(t) = \int_{-\infty}^{t} \underbrace{G\,e^{-(t-t')/\lambda}}_{G(t-t')}\,\dot{\gamma}(t')\,\mathrm{d}t'$$

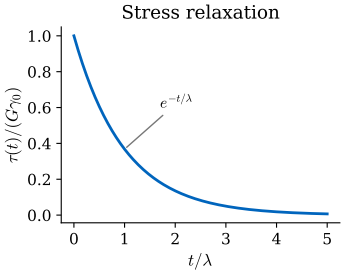
So the Maxwell model is the superposition principle with one exponential.

Two limits

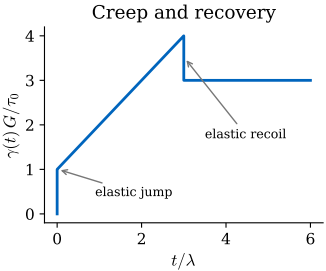
Fast forcing, $\lambda\dot{\tau} \gg \tau$: $\dot{\tau} = G\dot{\gamma}$, a solid.

Slow forcing, $\lambda\dot{\tau} \ll \tau$: $\tau = \eta\dot{\gamma}$, a liquid.

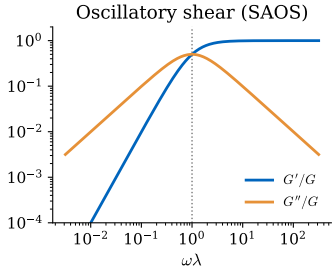
Maxwell: the three classic tests, solved exactly



γ_0 imposed at $t = 0$:
$\tau = G\gamma_0\, e^{-t/\lambda}$

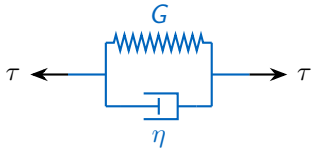


τ_0 imposed:
$\gamma = \tau_0/G + \tau_0 t/\eta$



$G' = G\,\omega^2\lambda^2/(1+\omega^2\lambda^2)$
$G'' = G\,\omega\lambda/(1+\omega^2\lambda^2)$
Crossover at $\omega\lambda = 1$ gives λ

The Kelvin–Voigt model: spring and dashpot in parallel



Same strain in both elements, stresses add:

$$\tau = G\gamma + \eta\dot{\gamma}$$

With the retardation time $\lambda_r = \eta/G$:

$$\lambda_r\,\dot{\gamma} + \gamma = \frac{\tau}{G}$$

Solving it, now for γ:

$$\gamma(t) = \frac{1}{\eta}\int_{-\infty}^{t} e^{-(t-t')/\lambda_r}\,\tau(t')\,\mathrm{d}t'$$

The mirror of Maxwell: the strain remembers the stress.

In the superposition form, $G(t) = G + 2\eta\,\delta(t)$. It never decays to zero, so $\eta_0 = \int_0^\infty G(s)\,\mathrm{d}s$ diverges: a solid.

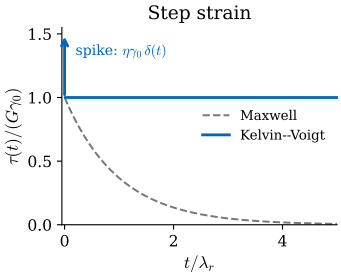
Two limits

Fast forcing, $\lambda_r\dot{\gamma} \gg \gamma$: $\tau = \eta\dot{\gamma}$, a liquid.

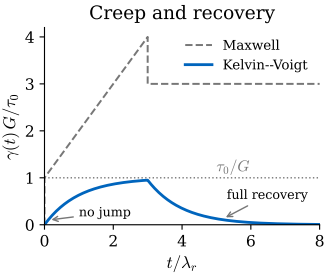
Slow forcing, $\lambda_r\dot{\gamma} \ll \gamma$: $\tau = G\gamma$, a solid.

The reverse of Maxwell.

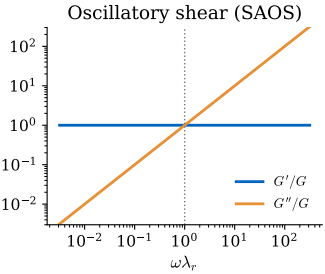
Kelvin–Voigt: the three classic tests, solved exactly



γ_0 imposed at $t = 0$:
$\tau = G\gamma_0$ for $t > 0$
No stress relaxation



τ_0 imposed:
$\gamma = (\tau_0/G)\left(1 - e^{-t/\lambda_r}\right)$
Removed at t_1:
$\gamma = \gamma(t_1)\, e^{-(t-t_1)/\lambda_r}$



$G' = G$, independent of ω
$G'' = \eta\omega = G\,\omega\lambda_r$
Crossover at $\omega\lambda_r = 1$ gives λ_r

Maxwell and Kelvin–Voigt, side by side

	Maxwell	**Kelvin–Voigt**
Elements	in series	in parallel
Shared, added	τ shared, $\dot{\gamma}$ add	γ shared, τ add
Time η/G	relaxation	retardation
Fast forcing	solid	liquid
Slow forcing	liquid	solid
Step strain	$\tau \to 0$	$\tau = G\gamma_0$
Creep, long times	flows, $\gamma \propto t$	saturates at τ_0/G
On unloading	keeps $\tau_0 t_1/\eta$	recovers everything
It is a	fluid that forgets	solid that responds slowly

We will meet it again: in the unyielded Saramito material, at each particle contact, and in the overdamped plug of a jammed packing.

What it cannot do

No instantaneous elasticity: a jump in strain needs an infinite stress. And it never flows. On its own it describes few real materials.

Add inertia

With a lumped mass m, $m\ddot{x} + \eta\dot{x} + Gx = 0$ is a damped oscillator, with a damping ratio

$$Z = \frac{\eta}{2\sqrt{mG}}$$

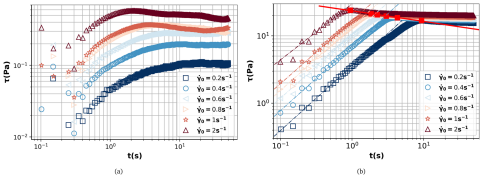
$Z < 1$: it rings. $Z > 1$: it creeps back without ringing.

Real fluids: many relaxation times

A single exponential rarely fits data. The generalised Maxwell model sums modes:

$$G(t) = \sum_k G_k \, e^{-t/\lambda_k}$$

- Our PAAm solution relaxes with a fast mode and a slower one (Traore *et al.*, 2015, Fig. 2)
- The longest time λ usually sets Wi
- Right: xanthan in start-up of shear, with an overshoot



Start-up of shear for (a) 0.2 % and (b) 1 % xanthan. Missi *et al.*, 2024, Fig. 8.

A *linear* model cannot overshoot: $\eta^+(t)$ grows monotonically. Overshoot is a non-linear signature.

Going 3D: why $\partial_t \boldsymbol{\tau}$ is not good enough

Naive tensorial Maxwell: $\lambda\, \partial_t \boldsymbol{\tau} + \boldsymbol{\tau} = 2\eta \mathbf{D}$.

Problem: rotate a stressed sample as a rigid body. $\mathbf{D} = 0$, and physically nothing should change, but the components of $\boldsymbol{\tau}$ change in the lab frame.

The fix: the time derivative must follow, stretch and rotate with the material. This is *material frame indifference*.

Upper-convected derivative

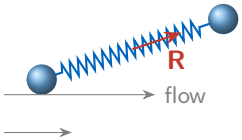
$$\overset{\nabla}{\boldsymbol{\tau}} = \partial_t \boldsymbol{\tau} + \mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u})\, \boldsymbol{\tau} - \boldsymbol{\tau}\, (\nabla \mathbf{u})^T$$

With $(\nabla \mathbf{u})_{ij} = \partial_j u_i$.

$\overset{\nabla}{\boldsymbol{\tau}} = 0$: stress simply carried and stretched by the flow, as if it were attached to material lines.

Key point: this term is nonlinear in the fields, and it is where N_1 comes from.

Where the model comes from: elastic dumbbells



Polymer coil = two beads and a Hookean spring, in a Newtonian solvent, with Brownian motion.

Average orientation: $\mathbf{A} = \langle \mathbf{RR} \rangle / R_0^2$

Kinetic theory gives a closed equation for $\mathbf{A}$:

$$\overset{\nabla}{\mathbf{A}} = -\frac{1}{\lambda}(\mathbf{A} - \mathbf{I}), \qquad \boldsymbol{\tau}_p = \frac{\eta_p}{\lambda}(\mathbf{A} - \mathbf{I})$$

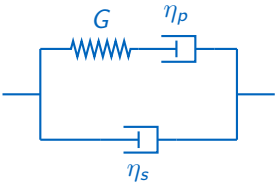
Using $\overset{\nabla}{\mathbf{I}} = -2\mathbf{D}$, this yields the upper-convected Maxwell model:

$$\lambda \overset{\nabla}{\boldsymbol{\tau}}_p + \boldsymbol{\tau}_p = 2\eta_p \mathbf{D}$$

Adding the solvent gives Oldroyd-B:

$$\boldsymbol{\tau} = \boldsymbol{\tau}_p + 2\eta_s \mathbf{D}$$

Oldroyd-B: the mechanical analogue and its numbers



Jeffreys analogue: a Maxwell element (polymer) in parallel with a dashpot (solvent).

- $\eta_0 = \eta_s + \eta_p$, viscosity ratio $\beta = \eta_s/\eta_0$
- $\lambda = \eta_p/G$
- Limits: $\beta = 0$: UCM; $\beta = 1$: Newtonian

Pencil-and-paper: steady shear

For $u_x = \dot{\gamma} y$, the UCM components give

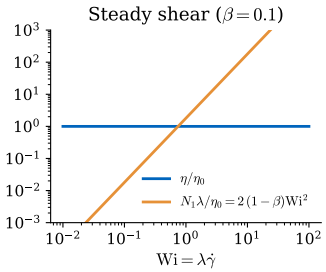
$$\tau_{yy} = 0$$
$$\tau_{xy} = \eta_p \dot{\gamma}$$
$$\tau_{xx} = 2\lambda\dot{\gamma}\,\tau_{xy} = 2\eta_p\lambda\dot{\gamma}^2$$

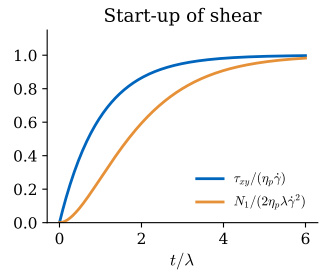
Thus **constant viscosity, and quadratic N_1:**

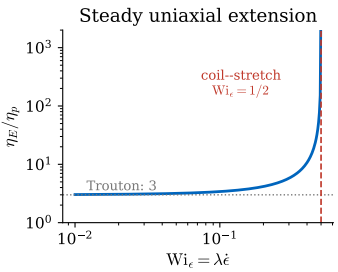
$$\eta = \eta_0, \quad N_1 = 2\eta_p\lambda\dot{\gamma}^2, \quad N_2 = 0$$

A good model of Boger fluids: elastic, not shear thinning.

Oldroyd-B: predictions







η constant, $N_1 \propto \mathrm{Wi}^2$

N_1 lags behind τ_{xy}

$$\frac{\eta_E}{\eta_p} = \frac{3}{(1-2\mathrm{Wi}_\epsilon)(1+\mathrm{Wi}_\epsilon)}$$

Weak spot: $\eta_E \to \infty$ at $\mathrm{Wi}_\epsilon = 1/2$, since the Hookean spring is infinitely extensible. Fixes: FENE-P (finite extensibility), Giesekus (anisotropic drag, gives shear thinning).

Viscoelasticity: what to keep

Takeaways

- Maxwell = one relaxation time: linear, 1D
- Frame indifference forces $\overset{\nabla}{\boldsymbol{\tau}}$
- Oldroyd-B: the minimal 3D viscoelastic fluid
- It explains N_1, the hoop stresses, and the Wi controlled instabilities behind elastic turbulence

Not captured

- Shear thinning: η is constant
- Yield stress: $\boldsymbol{\tau} \to 0$ as flow stops
- Finite extensibility
- Stress overshoot at start-up: components rise monotonically

Next: a material that does not relax to zero stress.

3. Introduction to viscoplasticity

Bingham and Herschel–Bulkley, and why they are not enough

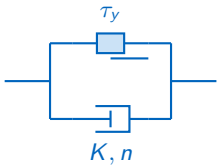
Bingham (1922) and Herschel–Bulkley (1926)

In simple shear:

$$\textbf{Bingham:} \quad \tau = \tau_y + \eta_{pl}\dot{\gamma}$$
$$\textbf{Herschel–Bulkley:} \quad \tau = \tau_y + K\dot{\gamma}^{\,n}$$
$$\text{both:} \quad \dot{\gamma} = 0 \ \text{ if } \ |\tau| \le \tau_y$$

Bingham is HB with $n = 1$; $\tau_y = 0$ recovers the power law.

Parameters: a single stress τ_y, and the viscous pair (K, n), fitted on a flow curve in the fluid regime.



Friction element in parallel with a (nonlinear) dashpot.

Below τ_y the friction element blocks everything: the material is perfectly rigid, with no elastic strain.

The 3D form, and a singular viscosity

Tensorial form with a von Mises criterion:

$$\boldsymbol{\tau} = \left(\frac{\tau_y}{|\dot{\gamma}|} + K|\dot{\gamma}|^{n-1} \right) 2\mathbf{D} \quad \text{if } |\boldsymbol{\tau}| > \tau_y$$

$$\mathbf{D} = 0 \quad \text{if } |\boldsymbol{\tau}| \leq \tau_y$$

with the invariants

$$|\boldsymbol{\tau}| = \sqrt{\tfrac{1}{2}\boldsymbol{\tau} : \boldsymbol{\tau}}, \qquad |\dot{\gamma}| = \sqrt{2\mathbf{D} : \mathbf{D}}$$

Note that $\boldsymbol{\tau} \parallel \mathbf{D}$: no normal stress differences in shear. Since $\mathrm{tr}\,\mathbf{D} = 0$, here $\boldsymbol{\tau}$ is traceless: it is its own deviatoric part.

A numerical difficulty

The effective viscosity diverges as $\dot{\gamma} \to 0$. Use augmented Lagrangian methods, or a regularisation (Papanastasiou):

$$\eta = K\dot{\gamma}^{n-1} + \tau_y \frac{1 - e^{-m\dot{\gamma}}}{\dot{\gamma}}$$

Large m approaches the ideal model, but the yield surface becomes blurred.

A worked example: plane channel flow

Steady flow in $|y| \leq h$. The momentum balance is *model-independent*:

$$|\tau_{xy}(y)| = \tau_w \frac{|y|}{h}, \qquad \tau_w = \frac{h\,\Delta P}{L}$$

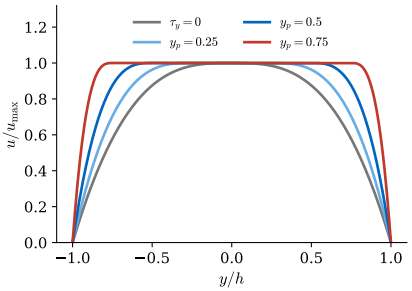
Plug where $|\tau_{xy}| < \tau_y$, so its half-width is

$$\frac{y_p}{h} = \frac{\tau_y}{\tau_w} \equiv \mathrm{Bn}$$

Integrating $K\dot{\gamma}^n = |\tau_{xy}| - \tau_y$ with no slip, for $\hat{y} = |y|/h > \mathrm{Bn}$:

$$u = \frac{n\,h}{n+1}\left(\frac{\tau_w}{K}\right)^{1/n}\left[(1-\mathrm{Bn})^{\frac{n+1}{n}} - (\hat{y}-\mathrm{Bn})^{\frac{n+1}{n}}\right]$$

No flow if $\mathrm{Bn} \geq 1$: a sharp threshold.



Herschel–Bulkley profiles, $n = 0.5$, at increasing $y_p/h = \mathrm{Bn}$.

A *complete* solution with three parameters: this is why HB is so widely used.

The "model" yield stress fluid picture

Herschel–Bulkley makes strong promises:

1. Yielding occurs at one stress, $\tau = \tau_y$
2. Below τ_y: perfectly rigid
3. Flow states are reversible with stress
4. No time dependence: no memory, no De

Carbopol gels were long considered to keep these promises: a "model" yield stress fluid. In the steady limit they hold up well (Younes *et al.*, 2020, Fig. 1).

The test

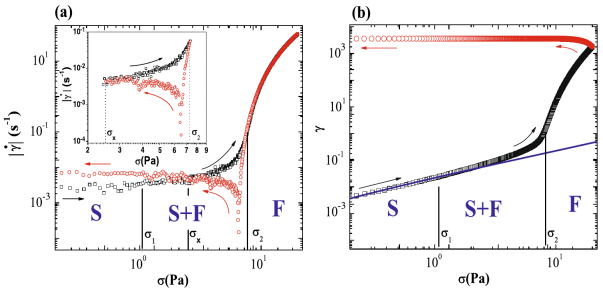
The steady flow curve is the *limit* $t_0 \to \infty$. So let us ask:

What happens if we load, then unload, at a finite rate?

Protocol: a stepped stress ramp, up then down, with a time t_0 per step.

Three regimes, not two: Putz & Burghelea (2009)

our work



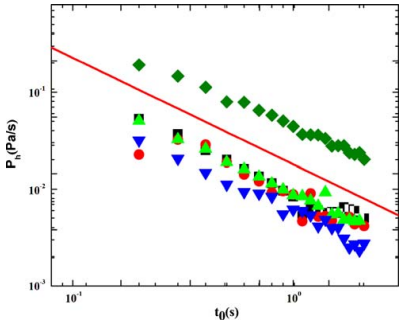
0.1 % Carbopol 940, controlled stress ramp, $t_0 = 0.2$ s. Squares: up; circles: down. (a) flow curve; (b) strain, fit $\gamma \approx 0.022\,\sigma$. Putz & Burghelea, Rheol. Acta 48 (2009), Fig. 3.

- S ($\sigma < \sigma_1$): Hookean, $\sigma = G\gamma$
- S+F ($\sigma_1 < \sigma < \sigma_2$): neither solid nor fluid
- F ($\sigma > \sigma_2$): HB and reversible

Hysteresis and a **cusp**: $\dot{\gamma}$ changes sign on unloading (elastic recoil). Energy lost in the loop: $W_h/W_e \approx 12$ here.

The hysteresis depends on the forcing time

our work



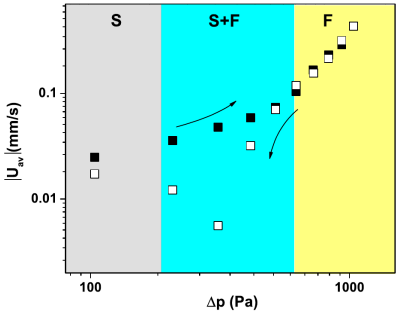
Power deficit P_h vs. time per stress value t_0, several concentrations and temperatures. Line: $P_h \propto t_0^{-1}$. Putz & Burghelea (2009), Fig. 11.

- Measured: $P_h \propto t_0^{-\alpha}$, with $\alpha = 0.98$–1.17
- Same for the Carbopol 940 gel of Younes *et al.* (2020): $A \propto t_0^{-1}$
- So $t_0 \to \infty$ recovers the steady, "model" results: no conflict with earlier studies
- Also seen in xanthan (Missi *et al.*, 2024, Fig. 5)

An apparent yield stress is meaningless unless the protocol is stated.

Not just a rheometer effect: hysteresis in a pipe

our work



Mean velocity vs. pressure drop, $t_0 = 4$ s. Full/empty: up/down. Poumaere *et al.*, JNNFM 205 (2014), Fig. 8.

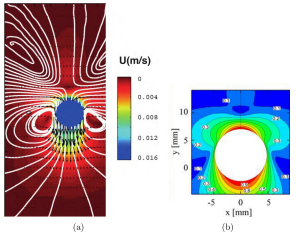
The same three regimes appear in a real flow:

- States overlap only in the fluid regime F
- Below it: hysteresis and elastic recoil
- With wall slip, a weak dependence on forcing time: $A \propto t_0^{-0.1}$

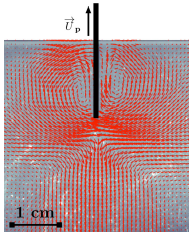
HB predicts zero flow below Δp_y: no plateau, no loop, no reversal. Elasticity *cannot be ignored* in unsteady yielding.

Simple flows at odds with the "model" picture

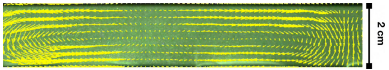
our work



Falling sphere, $\mathrm{Re} = 0.087$: fore–aft asymmetry and negative wake. (a) experiment; (b) simulation by Fraggedakis *et al.*



Withdrawn plate (Landau–Levich), $U_p = 3\,\mathrm{mm/s}$: negative wake again.

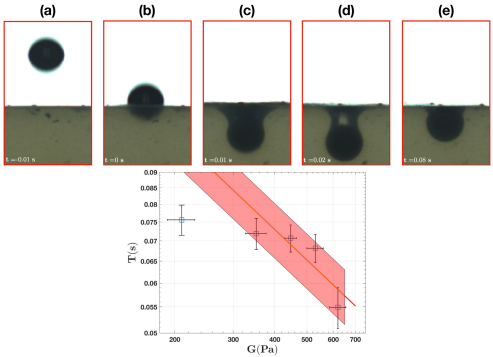


Rayleigh–Bénard: convection in a gel heated from below, contrary to stability studies of ideal yield stress fluids.

In creeping flow, an inelastic HB fluid gives a fore–aft symmetric field and no negative wake. These flows point to elasticity and a gradual transition near yield. Younes *et al.*, JNNFM 281 (2020), Figs. 2–4.

An elastic solid before it is a fluid: the stopping ball

our work



A sphere hits a Carbopol gel (top); oscillation period vs. modulus (bottom). Tarlet *et al.*, JNNFM 263 (2019), Figs. 3 and 6.

The ball does not just stop: it bounces, with damped oscillations, and rests at a finite depth.

The solid regime acts as a harmonic oscillator:

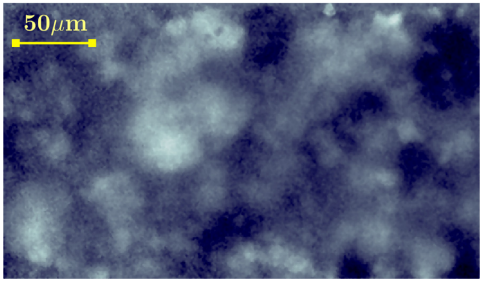
$$T = 2\pi R\sqrt{\frac{4\rho}{3\alpha}}\, G^{-1/2}$$

with α a constant of the model.

An HB material is rigid below τ_y: it *cannot* oscillate. Here G, measured independently by SAOS, sets the dynamics.

Is it damage? A look at the microstructure

our work



Carbopol gel covalently labelled with Rhodamine B: bright = particles, dark = solvent. Typical size $\sim 50\,\mu$m. Younes *et al.* (2020), Fig. 10.

In-situ micro-visualisation during a ramp shows:

- A highly structured gel
- Hysteresis is not caused by mechanical damage of the micro-gel
- Elastic effects in S and S+F contribute to it

"Model", thixotropic, or time-dependent?

A *time-dependent viscoplastic* material near yielding, tending to a "model" fluid only in the slow limit.

What a better model must contain

Observation	**Ingredient**	**Missing from HB:**
Plateau $\lvert\dot{\gamma}\rvert = \lvert\dot{\tau}\rvert/G$ below τ_y	Elastic solid	$\dot{\gamma} \equiv 0$ below τ_y
Recoil, reversal, overshoot	Stored elastic stress	No stress memory
Gradual transition S+F	Solid–fluid coexistence	Single sharp threshold
Hysteresis, $A \propto t_0^{-\alpha}$	An intrinsic time scale	No time in the equation
Normal stresses, negative wake	Tensorial elasticity	$\boldsymbol{\tau} \parallel \mathbf{D}$

Route A: structural kinetics

A variable for how much structure exists.
Putz–Burghelea; de Souza Mendes–Thompson

Route B: elasto-plastic switch

A sharp criterion, with elasticity added.
Saramito

4. Structural models

One scalar to describe the state of the microstructure

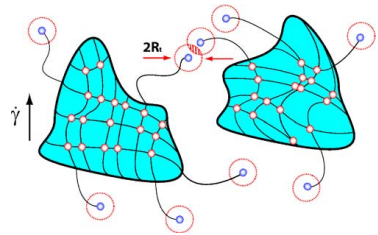
The recipe of a structural model

1. A scalar for the structure, usually in $[0, 1]$:
$1 =$ fully structured, $0 =$ fully broken

2. A kinetic equation:

$$\frac{\mathrm{D}\Phi}{\mathrm{D}t} = \underbrace{R_r}_{\text{recovery}} - \underbrace{R_d}_{\text{destruction}}$$

3. A constitutive law depending on Φ, with limits:
$\Phi \to 1$: elastic solid; $\Phi \to 0$: viscous fluid

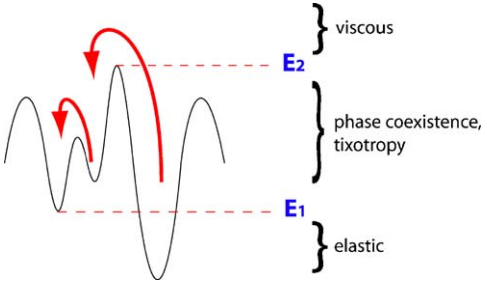


Shear breaks links between structural units; Brownian motion helps rebuild them. Putz & Burghelea (2009), Fig. 9.

R_r, R_d bring a time scale: structure lags behind the forcing.

Physical picture: an energy landscape with many minima

our work



Binding energies in a heterogeneous gel network (cartoon).
Putz & Burghelea (2009), Fig. 2.

- $E < E_1$: units deform intact $\Rightarrow$ elastic solid
- $E_1 < E < E_2$: some break and recombine differently $\Rightarrow$ coexistence
- $E > E_2$: all broken, rebinding negligible $\Rightarrow$ viscous

A suggested analogy

Hysteresis near a first-order transition, as in a ferromagnet. Only a formal analogy: a sheared gel is out of equilibrium.

The Putz–Burghelea model (2009)

our work

Kinetics of the solid fraction $\Phi = [S]$, with $\Gamma = \sigma/\sigma_2$:

$$\frac{\mathrm{d}\Phi}{\mathrm{d}t} = \underbrace{-K_d \Gamma \Phi}_{\text{destruction}} + \underbrace{K_r \left[1 - \tanh \frac{\Gamma - 1}{w}\right] \Phi(1-\Phi)}_{\text{recombination}} + \delta$$

Constitutive law, Maxwell type (as in Younes *et al.*, 2020):

$$\frac{\eta(\dot\gamma)}{G}\, \Phi\, \dot\tau + \tau = \eta(\dot\gamma)\, \dot\gamma, \qquad \eta = K|\dot\gamma|^{n-1} + \tau_y \frac{1 - e^{-m|\dot\gamma|}}{|\dot\gamma|}$$

Check the limits

$\Phi \to 1$, η large: $\dot\tau = G\dot\gamma$, Hooke

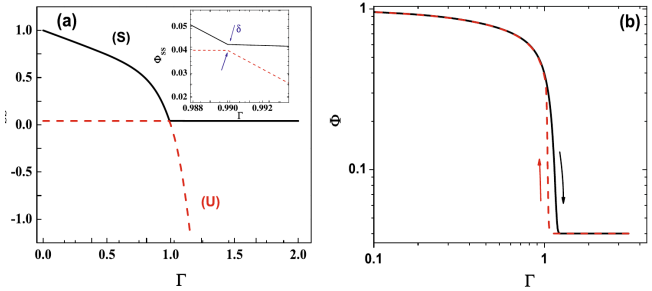
$\Phi \to 0$: $\tau = \eta(\dot\gamma)\dot\gamma$, HB

Stress imposed: $\Phi(t)$ decouples. Solve the ODE, then find $\dot\gamma$ algebraically.

$\Phi(1-\Phi)$: fluid units bind to existing solid blobs.

Steady states and the structural hysteresis

our work



(a) Steady states of the phase equation vs. Γ (S: stable, U: unstable); (b) Φ during a continuous stress ramp, for $K_r = 0.2$, $K_d = 0.3$, $w = 0.3$. Putz & Burghelea (2009), Fig. 12.

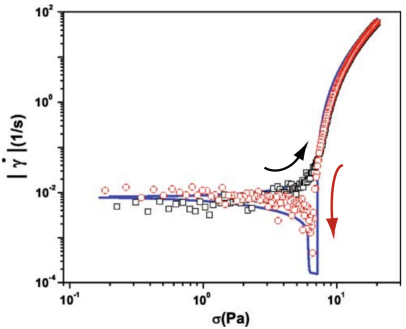
Let $g = K_d\Gamma$, and $f = K_r[1 - \tanh((\Gamma - 1)/w)]$. With $\delta = 0$:

$$\Phi = 0 \quad \text{or} \quad \Phi = 1 - \frac{g}{f}$$

- Exchange of stability when $g = f$
- δ regularises the crossing
- Slowing down near $g = f$ gives the lag

Comparison with experiment: stress ramps

our work



Symbols: experiment; lines: model. Putz & Burghelea (2009), Fig. 13.

Parameters:

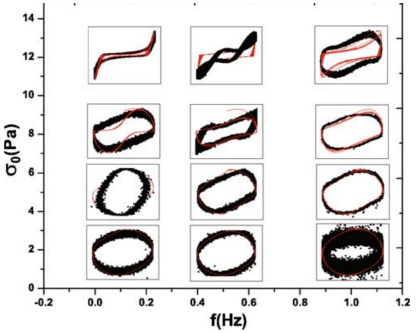
- G: fit of $\sigma = G\gamma$ in the solid regime
- K, n: HB fit in the fluid regime
- K_d, K_r, w: adjusted on the loop

A hint for Φ: $\Phi(\Gamma)$ resembles $G'(\sigma)$, so G' gauges how much solid remains.

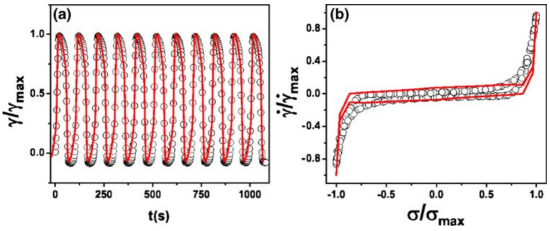
The plateau, the loop *and* the cusp are all reproduced.

The acid test: oscillatory flow, same parameters

our work



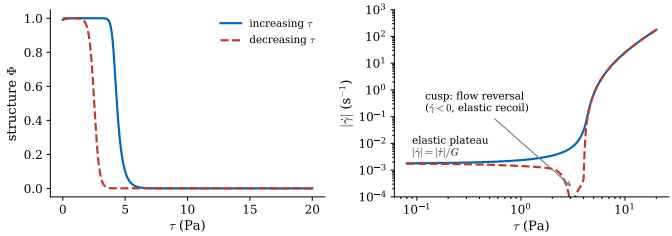
Pipkin space, 0.1 % Carbopol. Symbols: experiment; lines: model, parameters of Fig. 13. Putz & Burghelea (2009), Fig. 15.



$\sigma_0 = 8\,\mathrm{Pa}$, $f = 0.01\,\mathrm{Hz}$: strain series and Lissajous figure. Same paper, Fig. 14.

Pipkin space: one Lissajous figure per pair (f, σ_0), a "fingerprint" of the material. Agreement is good in S and S+F, and fair in F, where the neglected rheometer inertia matters.

Run the model yourself



Model: Younes *et al.* (2020), Eqs. 8–9, $\tau_y = 4\,\text{Pa}$, $G = 45\,\text{Pa}$, $K = 1.2$, $n = 0.5$, $k_r = 0.2$, $k_d = 0.1$, $w = 0.25$, $t_0 = 1\,\text{s}$. Illustrative parameters, **not a fit**. Script: `code/model_figures.py`.

Version of 2020

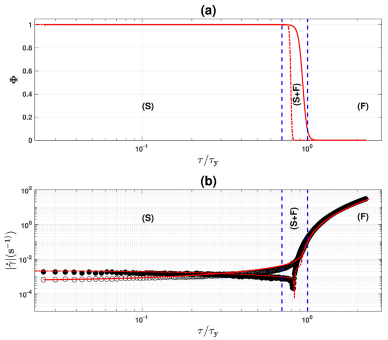
$$\dot{\Phi} = k_r \left[1 - \tanh \frac{\tau - \tau_y}{w}\right] \Phi(1-\Phi) - k_d \left[1 + \tanh \frac{\tau - \tau_y}{w}\right] \Phi + \delta$$

Destruction now acts only near and above yield.

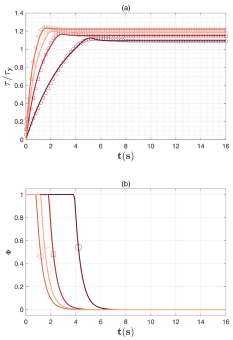
- S: $|\dot{\gamma}| = |\dot{\tau}|/G$
- Loop: Φ lags
- F: reversible HB

How simple can a scalar model be?

our work



Φ (a) and best fit, lines (b), of the measured ramp. Younes *et al.* (2020), Fig. 16.



Stress overshoot and Φ after a step in rate. Fig. 17.

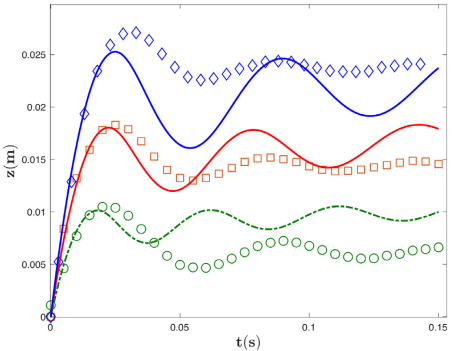
Tested on a Carbopol 940 gel:

- Coussot's model (2002), with no elasticity, fails: no overshoot
- The elastic structural model captures ramps *and* overshoot

Simple scalar models work "as long as they are not too simple".

Further uses and foundations

our work



Stopping ball at three release heights. Symbols: experiment; lines: model. Tarlet *et al.* (2019), Fig. 5.

Applied to flows: stopping sphere (Tarlet *et al.*, 2019); unsteady pipe flow with slip laws (Moyers-González *et al.*, 2022).

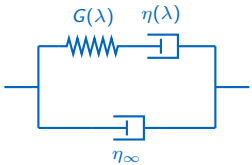
Microscopic basis: a Gibbs random field of interacting units (Sainudiin *et al.*, 2015), whose ODE approximation agrees very well with Putz–Burghelea (Burghelea *et al.*, 2017).

Limitation: as written, a scalar model suited to rheometric flows. Complex flows need a tensorial form.

The PUC-Rio school: de Souza Mendes & Thompson

A *tensorial, frame-indifferent* line of models:

- de Souza Mendes (2009): equations for stress and structure. Flow curve exact by construction; overshoot, viscosity bifurcation, non-monotonic curves
- de Souza Mendes (2011): adds elasticity: a thixotropic elasto-viscoplastic (TEVP) model
- de Souza Mendes & Thompson (2012, 2013): a critical overview, then a unified approach for yield-stress and apparent yield-stress materials
- Thompson *et al.* (2018): "the yield stress tensor"



Analogue of the 2011 model: structure-dependent Maxwell element in parallel with a Newtonian one.

Familiar? It is Jeffreys / Oldroyd-B with parameters depending on λ.

The TEVP model of de Souza Mendes (2011)

Total stress $\boldsymbol{\tau} + 2\eta_\infty \mathbf{D}$, with

$$\boldsymbol{\tau} + \theta(\lambda)\overset{\nabla}{\boldsymbol{\tau}} = 2\eta(\lambda)\mathbf{D}, \quad \theta = \frac{\eta(\lambda)}{G(\lambda)}$$

$$\eta = \eta_\infty \left(\frac{\eta_0}{\eta_\infty}\right)^\lambda, \quad G = G_0\, e^{m(1/\lambda - 1)}$$

Kinetics, relaxing towards an equilibrium $\lambda_{eq}(\dot\gamma)$:

$$\frac{\mathrm{D}\lambda}{\mathrm{D}t} = \frac{1}{t_{eq}}\left[(1-\lambda) - (1-\lambda_{eq})\frac{\lambda}{\lambda_{eq}}\frac{\tau}{\eta(\lambda)\dot\gamma}\right]$$

Written as in Frey, Link, Ferreira, Naccache & de Souza Mendes, Proc. SBMAC 3 (2015). Exponents on the kinetic terms may differ in other versions of the model.

Read the limits

$\lambda \to 1$: $G = G_0, \eta = \eta_0$, long θ: solid-like

$\lambda \to 0$: $G \to \infty, \theta \to 0$: Newtonian, η_∞

$\lambda = \lambda_{eq}$, $\tau = \eta\dot\gamma$: kinetics vanish, on the steady flow curve $\eta_{eq}(\dot\gamma)$

The role of the flow curve, and an important idea

λ_{eq} follows from the *measured* steady viscosity:

$$\lambda_{eq} = \frac{\ln \eta_{eq}(\dot{\gamma}) - \ln \eta_\infty}{\ln \eta_0 - \ln \eta_\infty}$$

for example with the SMD function (de Souza Mendes & Dutra, 2004):

$$\eta_{eq} = \left[1 - e^{-\frac{\eta_0 \dot{\gamma}}{\tau_0}}\right] \left[\frac{\tau_0 - \tau_{0d}}{\dot{\gamma}} e^{-\frac{\dot{\gamma}}{\dot{\gamma}_{0d}}} + \frac{\tau_{0d}}{\dot{\gamma}} + K\dot{\gamma}^{n-1}\right] + \eta_\infty$$

τ_0, τ_{0d}: static and dynamic yield stress. If $\tau_0 > \tau_{0d}$, a non-monotonic curve is possible.

True or apparent yield stress?

With a large finite η_0, the structured state creeps very slowly: an *apparent* yield stress. The 2013 approach treats both classes.

The price

About 10 parameters: $\eta_0, \eta_\infty, G_0, m, t_{eq}, \tau_0, \tau_{0d}, \dot{\gamma}_{0d}, K, n$. But it is frame-indifferent: usable in complex flows.

Two structural philosophies, side by side

	Putz–Burghelea	**de Souza Mendes–Thompson**
Structure variable	Φ, solid fraction	λ, structuring level
Stress law	Scalar, Maxwell type	Tensorial, Jeffreys / Oldroyd-B type
Time scales	$1/K_r$, $1/K_d$; η/G	t_{eq}; $\theta(\lambda)$
Driven by	Imposed stress	Stress vs. equilibrium value
Steady flow curve	HB in the limit $\Phi \to 0$	Exact by construction
Designed for	Carbopol ramps, LAOS	General structured fluids

Shared difficulty: Φ or λ are hard to measure, and several parameters must be adjusted. Hence validation on an independent flow, such as the Pipkin space.

5. A particle view of yielding

No structure parameter: only springs, dashpots and Newton's law

From postulating Φ to computing the material

our work

Parts 3–4: **Postulate** τ_y, G, K, n and a kinetics for Φ → Compute the flow → **Compare** with measurements

This part: **Postulate** one contact: a spring and a dashpot → Integrate Newton's law for each disk → **Measure** σ_y, G, K, n, time scales, thresholds

A deliberately poor material

No attraction, no bonds, no ageing, no temperature. There is nothing to break and rebuild: it cannot be thixotropic in the usual sense.

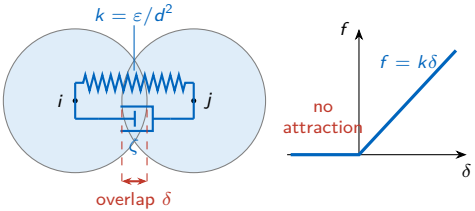
So the question becomes

Which yielding behaviours survive without any structural kinetics, and which ones really call for a Φ?

Same test sequence as for the Saramito model in Burghelea & Moyers-González (2025). Throughout this part σ is the shear stress, written τ elsewhere in the tutorial.

Ingredient 1: a single contact

our work



$$m_i \ddot{\mathbf{r}}_i = \sum_j \Big[\underbrace{f_{ij}^{\mathrm{el}}\, \hat{\mathbf{n}}_{ij}}_{\text{spring}} \underbrace{-\zeta(\mathbf{v}_i - \mathbf{v}_j)}_{\text{dashpot}} \Big] + m_i g\, \hat{\mathbf{x}}$$

Recognise it? Close to a Kelvin–Voigt element at each contact; but the spring acts only along the line of centres, while the dashpot also resists sliding.

One number for the damping. Isolated pair: $\mu\ddot{x} + \zeta\dot{x} + kx = 0$, so

$$Z = \frac{\zeta}{2\sqrt{k\mu}} = \frac{\zeta}{\sqrt{2}}$$

$Z = 0.71,\ 2.1,\ 7.1$: from a contact that just rings to one strongly overdamped.

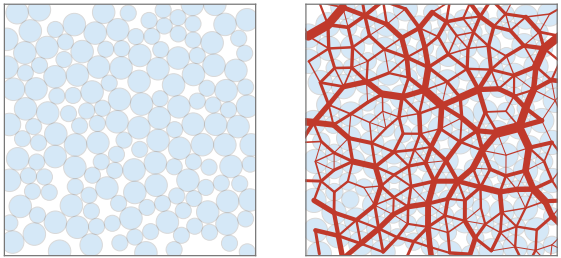
No friction, no rotation of the disks.

Dashpot on the full relative velocity, as in Durian's bubble model (1995). A drag against a fixed background would absorb the force locally, and no stress would reach the walls.

Ingredient 2: squeeze the disks, get a solid

our work

$\phi = 0.76 < \phi_J$: no contacts, no solid $\phi = 0.88 > \phi_J$: a force network



Computed for this tutorial: 160 disks relaxed to mechanical equilibrium. Red lines join disks in contact; their width follows the contact force.

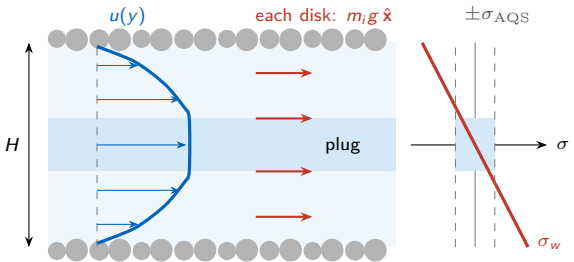
- Mixture of diameters 1 and 1.4: it cannot crystallise
- Above jamming, $\phi_J = 0.843$, the disks must overlap
- The stress is carried by a force network, very uneven

A solid with no bonds: its rigidity comes from crowding alone.

Here $\phi = 0.88$, with $G = 0.054$ and $p = 0.012$ in units of ε/d^2.

Ingredient 3: a channel flow without a pressure

our work



frozen, rough walls, cut from the same packing

Momentum balance, for any material:

$$\sigma_{xy}(y) = \rho g\,(y_c - y)$$

$$\sigma_w = \frac{\rho g H}{2}$$

A single control parameter:

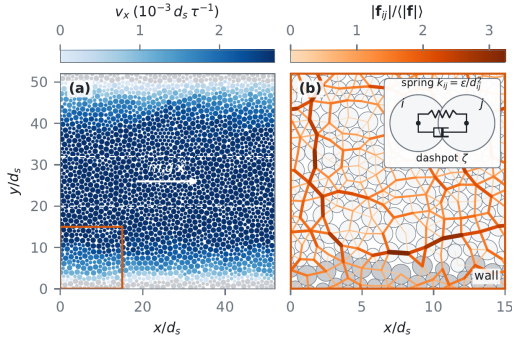
$$S = \frac{\sigma_w}{\sigma_{\mathrm{AQS}}}$$

S^{-1} plays the role of a Bingham number. σ_{AQS}: see the next slides.

$H \approx 48\,d$, 2048 disks of which 1891 mobile; periodic along x. Compare with the worked channel of Part 3.

What it looks like: steady flow at $S = 4$

our work



(a) Disks coloured by streamwise velocity; dashed lines: $|\sigma_{xy}| = \sigma_{\mathrm{AQS}}$. (b) Zoom on the frame of (a), contact forces near the wall; inset: one contact. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 1.

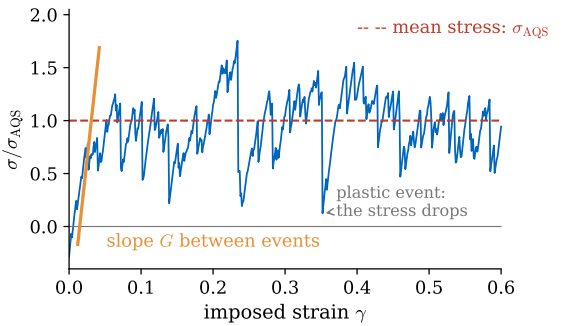
How to read it:

- Between the dashed lines the stress is below σ_{AQS}: the dark, nearly uniform plug
- Near the walls: the sheared layers
- The force network is not frozen: it keeps breaking and re-forming

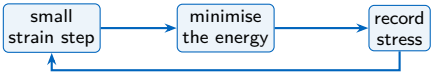
The measured stress follows the momentum balance to within about 1 %: the slab really is a channel flow.

Measuring the yield stress with no time at all

our work



Computed for this tutorial: one packing of 1024 disks, strain steps of 10^{-3}, up to $\gamma = 0.6$. Elastic slope 4 % above the G of the manuscript, mean stress 10 % above its σ_{AQS}. It starts below zero: the quench freezes in a residual stress.



Athermal quasistatic shear (AQS): no time, no inertia, no damping. Only the geometry of the energy landscape.

$\sigma_{\mathrm{AQS}} = 8.26 \times 10^{-4}\, \varepsilon/d^2$, from packing to packing within a few per cent. The stress unit from now on.

Yield strain: $\sigma_{\mathrm{AQS}}/G = 0.015$, small.

Stress plateau over strains 0.5–3. Linear in the distance to jamming, as harmonic contacts demand: $\sigma_{\mathrm{AQS}} d^2/\varepsilon = 0.021\,(\phi - \phi_J)$.

Who is overdamped? Contacts, modes and clocks

our work

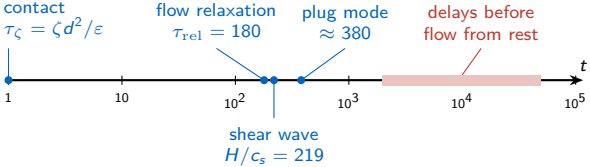


Z	0.71	2.1	7.1
Overdamped eigenvalues	77 %	98 %	99.9 %
Oscillating modes	866	67	3

Exact linear dynamics of the quenched channel, 3766 degrees of freedom. Manuscript, Table I.

A surprise: a contact with $Z < 1$ still rings, but every channel-scale mode is overdamped. Many contacts, each damping the relative motion.

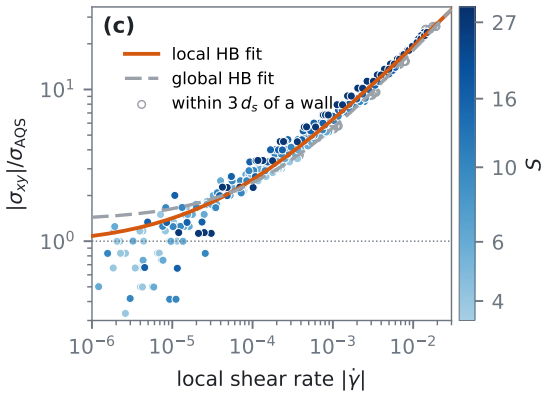
The clocks of the problem (at $Z = 0.71$):



Time unit $\sqrt{md^2/\varepsilon}$. The delays span 2×10^3 to 5×10^4.

Steady flow: Herschel–Bulkley comes out by itself

our work



Local stress (momentum balance) against local shear rate, from ten profile runs, $S = 4$–27. Filled: more than $3\,d$ from a wall. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 2c.

Nothing in the contact law says "Herschel–Bulkley", and yet

$$\sigma = \sigma_y + K\dot{\gamma}^{\,n}$$
$$\sigma_y = 0.94\,\sigma_{\mathrm{AQS}},\ n = 0.53,\ K = 0.171$$

The local yield stress is the quasistatic one, and all the runs nearly collapse: the flowing states are local.

The trap of the global fit

Fitting $Q(\sigma_w)$ with a no-slip HB profile gives $\sigma_y = 1.37\,\sigma_{\mathrm{AQS}}$, or anywhere in 1.15–1.70, depending on the stress range, always with a good χ^2. Wall slip ($\sim 12\,\%$) and the σ_y–n correlation are to blame.

The flow curve, written with microscopic parameters

our work

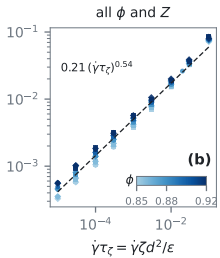
Dimensional analysis first. With stress unit ε/d^2 and viscous time $\tau_\zeta = \zeta d^2/\varepsilon$: $\sigma d^2/\varepsilon = F(\phi, \dot\gamma\tau_\zeta, Z)$.

Then simulations (165 runs, homogeneous shear) give F:

$$\sigma = \frac{\varepsilon}{d^2}\left[\underbrace{0.021\,(\phi-\phi_J)}_{\text{jamming}} + \underbrace{0.21\left(\dot\gamma\zeta d^2/\varepsilon\right)^n}_{\text{contact dissipation}}\right]$$

with $n \approx 0.58 - 0.9\,(\phi - \phi_J)$.

- σ_y does not know about the damping
- $K \propto \zeta^n$, verified over a factor 33
- Z drops out: inertia changes the stress by less than 5 % up to an inertial number ≈ 0.03

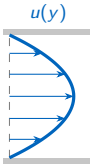


Excess stress for all ϕ and Z: one power law. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 3b.

The law of dissipation matters: a drag against the affine flow instead of contact dashpots gives $n = 0.39$–0.45 and a five times smaller k.

Below yield: an elastic plug, computed on the blackboard

our work



A clamped slab, loaded by the body force:

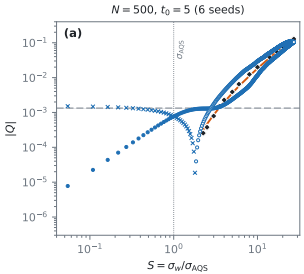
$$G u'' = -\rho g, \quad u\left(\pm \tfrac{H}{2}\right) = 0$$

Average the parabola:

$$\bar{u} = \frac{\rho g H^2}{12\, G} = \frac{\sigma_w H}{6G}$$

If the stress is ramped, a mean velocity is imposed:

$$Q_{\mathrm{el}} = \frac{H \dot{\sigma}_w}{6G} \qquad \text{no free parameter}$$

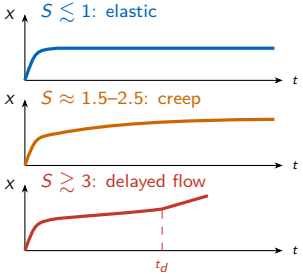


Filled: up; open: down; crosses: backflow. Dashed: Q_{el}, nothing fitted. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 6a.

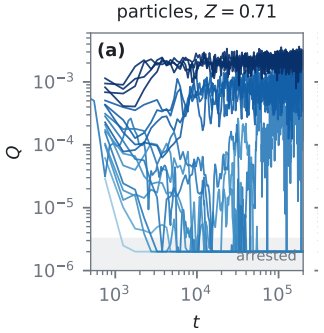
Up: $Q = 0.97\, Q_{\mathrm{el}}$ until $S \approx 3$. **Down:** $Q = -1.12\, Q_{\mathrm{el}}$: the plug flows back. The plateau and the cusp of Part 3.

Load it and wait: three fates

our work



Sketches. X: plug displacement.



$Q(t)$ at $Z = 0.71$; light to dark: $S = 0.7$ to 4; one line per seed; arrested runs on the floor. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 4a.

Recovered on unloading

$S \leq 1$: 86–100 %; $S = 2$: 60 %; $S = 3$: 6 % (longest hold)

Fraction flowing, $Z = 0.71$

$S = 2$: 0/4; $S = 2.5$: 1/4; $S = 3$: 4/4

Delays

5×10^4 near threshold, 2×10^3 at $S = 4$. Scaled by ζ, the three dampings fall in the same range: not an inertial time.

The same flow, at $S = 2.5$: solid or fluid, it depends on the sample.

Hysteresis with no structure: a toy lag

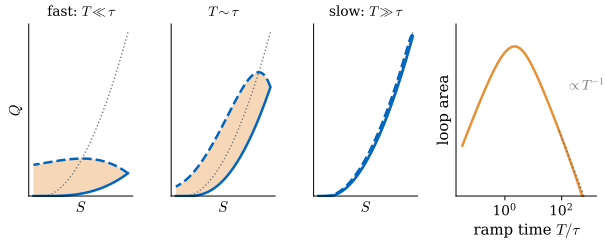
our work

A velocity field cannot adapt at once: momentum has to diffuse across the channel. After a small stress change,

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{Q_{ss}(\sigma_w) - Q}{\tau(\sigma_w)}, \qquad \tau = \frac{\rho H^2}{\pi^2 \eta}$$

with $\eta = (H/6)\,\mathrm{d}\sigma_w/\mathrm{d}Q_{ss}$, the tangent viscosity.

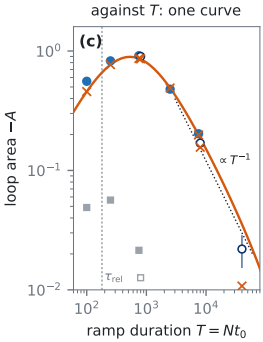
There is no Φ here, and nothing to fit: only ρ and the steady flow curve. If $\rho \to 0$ (Stokes), the channel follows the stress instantly and the loop vanishes.



Toy computed for this tutorial, constant τ. Solid: up; dashed: down; dotted: steady curve.

The measured loop obeys the lag

our work



Signed loop area against ramp duration, two protocols. Line: lag model, *no fit*. Within 2–18 % for $T = 100$–8000, half the area for the slowest ramp. Grey squares: near-yield window only. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 8c.

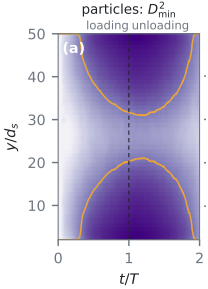
- Plot against the ramp duration $T = Nt_0$, not t_0: two protocols collapse
- Peak at $T \approx 4\,\tau_{\mathrm{rel}}$, then $A \propto T^{-1}$
- 87–100 % of the area lies in the fluidised regime, $S \geq 3$: this loop is *not* about yielding
- The shape of Divoux *et al.* (2013), here without thixotropy

A loop does not prove structure

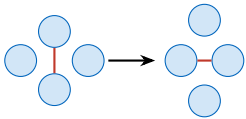
But do not over-read it either: in a 1 mm rheometer gap with $\eta \gtrsim 1\,\mathrm{Pa\,s}$, $\rho H^2/\eta \lesssim 1\,\mathrm{ms}$: too short to explain the Carbopol loops of Part 3, at $t_0 \geq 0.2\,\mathrm{s}$. **Estimate the time scale first.**

The packing remembers it has flowed

our work



$D^2_{\min}$ along a fast ramp: white 10^{-6}, dark 10^{-2}; dashed: end of loading; orange: plug edges. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 10a.



Neighbours change: a rearrangement that no local strain can describe.

$$D^2_{\min} = \min_{\mathbf{F}} \sum_j \left| \mathbf{r}_{ij}(t + \Delta t) - \mathbf{F}\,\mathbf{r}_{ij}(t) \right|^2$$

Falk & Langer (1998): the part of the motion that is not affine. Large where the material is plastic.

Loading: activity starts at the walls and invades late.

Unloading: it decays slowly; the plug re-forms long after the stress fell.

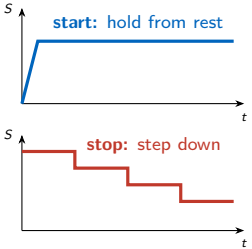
At equal S, unloading against loading: $\times 40$–50 at $S = 2$–3, $\times 350$ at $S = 1$; still $\times 10$ in a slow ramp.

Recoil after flow: $1.6\times$ the elastic value.

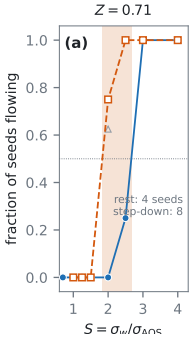
With no Φ in the equations, the state of the packing still depends on its history. Could such a field be what Φ means? Open.

Not one threshold: two

our work



Sketches. Each packing is one "seed" of the disorder; the threshold is where half of the seeds flow.



Fraction of seeds flowing. Filled: from rest; open: after step-down; shaded: the gap. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 11a.

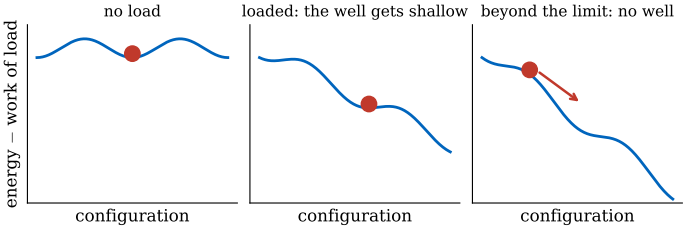
Z	Start	Stop	Gap
0.71	2.67	1.83	0.83
2.1	2.33	1.83	0.50
7.1	2.38	2.17	0.21

Units of σ_{AQS}; $H = 48\,d$. Resolved at $Z = 0.71$. At $Z = 7.1$ the brackets overlap: consistent with zero. A threshold is a distribution over packings, known to a grid step of 0.25–0.5.

The puzzle: both lie two to three times above the yield stress measured a few slides ago. Why? And what is the gap?

The start: losing stability, not waiting

our work



Cartoon of a tilted landscape, in the spirit of the many-minima picture of Part 4. Each load tilts it further; at a critical tilt, the minimum vanishes.

The test: load the same channel by AQS, with no time, no inertia, no damping. Stability is lost at

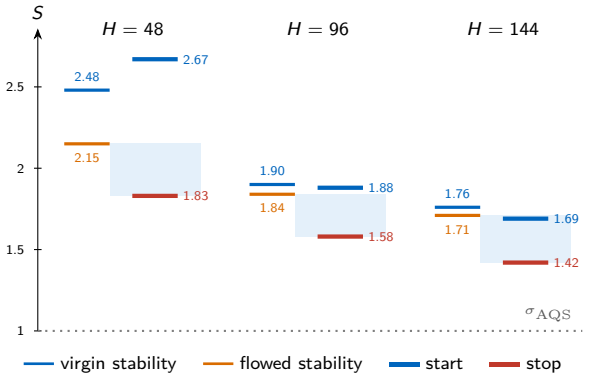
$$S_{\mathrm{AQS}} = 2.48 \pm 0.28, \text{ against } S_{\mathrm{start}} = 2.67.$$

Seed by seed, it predicts 57 of 64 holds at $H = 96$ and 144.

The delays are a slow escape past a mechanical instability, on the overdamped time $\zeta d^2/\varepsilon$. They are not waiting times for an activated event: there is no temperature.

The stop: a flowing packing does not find the stable states

our work



Left of each pair: AQS, no dynamics. Right: dynamic runs, $Z = 0.71$. Shaded: flowed stability minus stop. Eight seeds. Manuscript, Table V.

The gap has **three parts:**

1. start − virgin: ≈ 0 within the grid

2. virgin − flowed: flow weakens the packing, closes with H

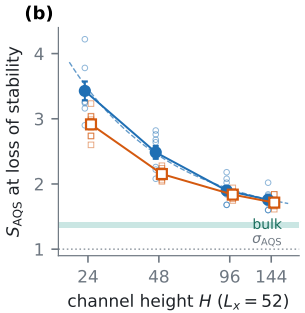
3. flowed − stop: ≈ 0.3 at all H

Part 3 is dynamic. A flowing packing keeps flowing where stable configurations exist: it does not find them. Like a ball rolling over wells where it could have rested.

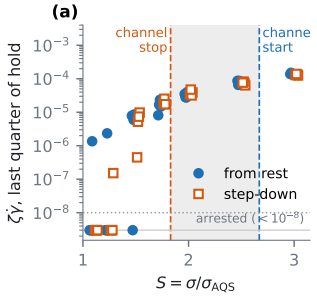
The role of inertia is suggestive, not proved: part 3 vanishes at $Z = 7.1$ at $H = 48$, but the bulk shows no inertial effect.

Is it the material, or the channel?

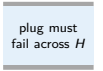
our work



Stability limit against H: virgin (filled), flowed (open). Dashed: $S_\infty + a/H$. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 12b.



No walls, stress imposed, one point per packing: from rest and after step-down coincide. Dashed: the channel values at $H = 48$. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 12a.



Channel: S_{AQS} falls as $1/H$; $(S - S_\infty)H$ is constant. A fixed excess force has to cross the channel.

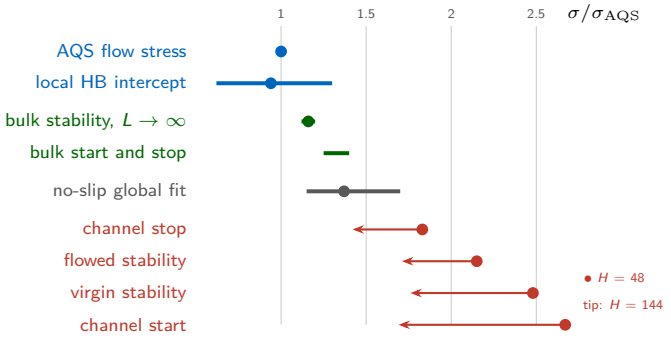
No fixed length: the wall layer above σ_{AQS} at the start stress is 16, 24, 32 d.

Bulk: start 1.38, stop 1.33, at $N = 1024$. No resolved gap, at any box size.

The flowing states stay local (Q/H constant for $H = 48$–144). Where they end depends on the size.

So how many "yield stresses" did we meet?

our work



Dots: best value. Bars: range over fits or sizes. Bulk start and stop: dynamic bracket in the largest box resolved, $L = 73.6\,d$.

Material: what the flow curve needs.

No walls: a dynamic threshold, falling with box size.

Fit: an artefact.

Channel: geometry, protocol and history.

Before you quote a yield stress, say: which protocol, which geometry, which history?

Back to the structure parameter: what is left for Φ?

our work

Observed in the packing	What produces it	Structure variable?
Elastic branches, backflow	G and the stress rate: $Q_{\mathrm{el}} = H\dot{\sigma}_w/6G$	not needed
HB flow curve	Contact dissipation near jamming	not needed
Ramp loop, peak at $\approx 4\tau_{\mathrm{rel}}$	Momentum diffusion: a lag	not needed
High start threshold, delays	Loss of stability, finite size	not needed
Stop ≈ 0.3 below stability	Flow state: stable states not found	a stop condition tied to the flow
Activity after flow; recoil above elastic; solid relaxation differs on unloading	Flow history of the packing	**called for**

The message for Part 4

Much of what we would credit to Φ exists without it. Test the structure-free explanations first, then add a structure variable where they fail.

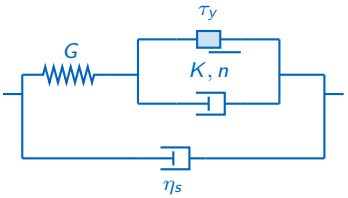
What this does not say

2D, frictionless, repulsive disks, by design. Carbopol has more: whether a persistent loop needs attraction or structural memory is **open**, and the comparison with its unsteady flows is still to be done.

6. The Saramito model

No structure variable: just a yield criterion added to Oldroyd-B

The idea: merge Oldroyd-B with Herschel–Bulkley



After Saramito (2009): spring in series with friction || HB dashpot, in parallel with the solvent.

Friction blocks: spring + solvent, a Kelvin–Voigt solid.

Friction slides: a viscoelastic fluid.

1D:

$$\frac{\dot\tau}{G} + \max\left(0, \frac{|\tau| - \tau_y}{K|\tau|^n}\right)^{1/n} \tau = \dot\gamma,$$

$$\sigma = \tau + \eta_s \dot\gamma$$

3D:

$$\frac{1}{G}\overset{\nabla}{\boldsymbol\tau}_p + S\,\boldsymbol\tau_p = 2\mathbf{D}, \quad S = \max\left(0, \frac{|\boldsymbol\tau_p^d| - \tau_y}{K|\boldsymbol\tau_p^d|^n}\right)^{1/n}$$

Deviatoric part ($N = 2$ or 3, the space dimension):

$$\boldsymbol\tau^d = \boldsymbol\tau - \tfrac{1}{N}(\operatorname{tr}\boldsymbol\tau)\mathbf{I}, \qquad |\boldsymbol\tau^d| = \sqrt{\tfrac{1}{2}\boldsymbol\tau^d : \boldsymbol\tau^d} \text{ (von Mises)}$$

Unlike in HB, $\boldsymbol\tau_p$ is not traceless: its isotropic part must not cause yielding. The model satisfies the second law, built from convex energy and dissipation potentials.

Pedagogical check: every limit we know

Limit	Result
$\tau_y = 0,\ n = 1$	$S = 1/K$: Oldroyd-B, $\eta_p = K,\ \lambda = K/G$
$G \to \infty$	$S\boldsymbol{\tau}_p = 2\mathbf{D}$: Herschel–Bulkley
$\lvert\boldsymbol{\tau}_p^d\rvert \le \tau_y$	$S = 0,\ \overset{\nabla}{\boldsymbol{\tau}}_p = 2G\mathbf{D}$: elastic solid
$n = 1$	Bingham version (Saramito, 2007)

A stress-dependent relaxation time:

$$\lambda_{\text{eff}} = \frac{1}{G\,S} \quad \Rightarrow \quad \mathrm{Wi} = \frac{\dot{\gamma}}{G\,S}$$

from ∞ when unyielded, to small when strongly sheared.

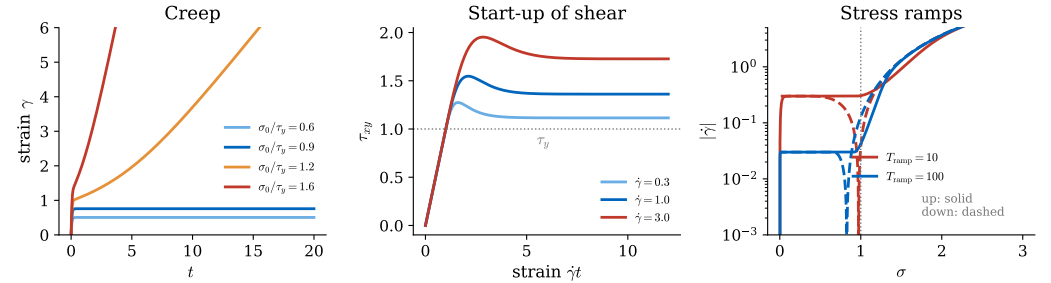
Only 5 parameters

G, τ_y, K, n, η_s. SAOS gives G; a steady flow curve gives τ_y, K, n.

What was removed

No structure variable and no kinetics. Yielding is decided by a sharp criterion.

Solving it in simple shear



Model, homogeneous shear, full tensorial equations: $G = \tau_y = K = 1$, $n = 0.5$, $\eta_s = 0.05$, no inertia. Script: `code/model_figures.py`.

Bounded strain below yield (solid); unbounded above (fluid).

Overshoot comes from the tensorial term: N_1 enters $|\boldsymbol{\tau}^d|$.

Elastic plateau $|\dot{\gamma}| = |\dot{\sigma}|/G$; recoil on the way down; rate-dependent loop.

A subtlety: elastic loading lowers the apparent yield stress

Unyielded, $\overset{\nabla}{\boldsymbol{\tau}} = 2G\mathbf{D}$ integrates in simple shear to

$$\tau_{xy} = G\gamma, \qquad N_1 = \tau_{xx} = \tau_{xy}^2/G \quad \text{(neo-Hookean)}$$

With $N = 2$ and $\tau_{yy} = 0$: $\tau_{xx}^d = -\tau_{yy}^d = \tau_{xx}/2$, so $|\boldsymbol{\tau}^d|^2 = \tau_{xy}^2 + \tau_{xx}^2/4$. Yielding occurs at

$$\tau^* \sqrt{1 + \frac{(\tau^*)^2}{4G^2}} = \tau_y$$

so $\tau^* \simeq \tau_y$ only if $\tau_y \ll G$. In steady *flow*, $N_1 = 2\tau_{xy}^2/G$: a different threshold on the way back.

Numbers

Slow ramp, $\tau_y = G = 1$:
formula 0.910; simulation 0.911

Channel, $\tau_y = 2, G = 1, \Delta P = 10$:
$W_p = 2\tau^*/\Delta P = 0.314$;
paper's validation run: 0.3147

Derived for this tutorial. It matters when τ_y/G is not small.

Test: unsteady plane channel flow

our work

Burghelea & Moyers-González, TCFD 39 (2025):

- Herschel–Bulkley Saramito model (SRM-HB)
- No slip; Basilisk-C, adaptive mesh
- $\eta_s = K = 1$, $n = 0.5$, usually $G = 1$; $\mathrm{Re}^{\max} = 0.3$
- $\mathrm{Bn} = 2L\tau_y/(W\Delta P)$

Protocols, as in the rheometer: creep; up/down pressure ramps; large amplitude oscillations.

Why a channel?

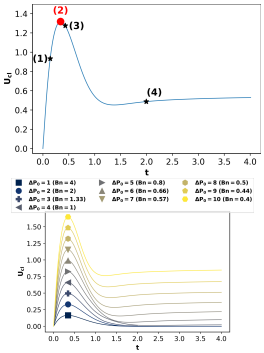
A *bridge* between viscometric and complex flows: simple kinematics, inhomogeneous stress, and comparable to Carbopol pipe experiments.

Why this model?

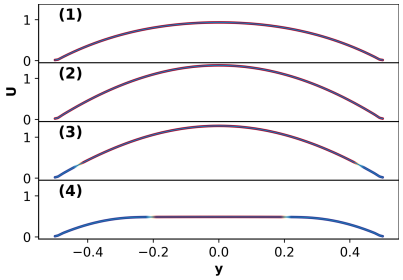
It came out ahead in a comparison of EVP models (Fraggedakis *et al.*, 2016), and its tensorial form suits non-rheometric flows.

Creep: overshoot, then yielding from the walls

our work



$\Delta P_0 = 8$ (top); several ΔP_0 (bottom). Figs. 2–3.



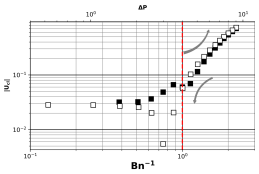
Profiles at the instants (1)–(4). Red: unyielded; blue: yielded. Burghelea & Moyers-González (2025), Fig. 3.

- (1)–(2): unyielded, yet curved profile: elastic solid, not rigid
- (3): yields near the walls
- (4): flat plug, HB-like

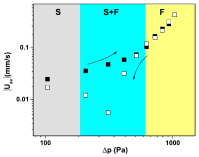
Overshoot weakens as G grows: an elastic effect.

Pressure ramps: hysteresis and flow reversal

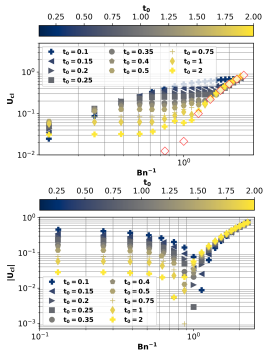
our work



SRM-HB, $t_0 = 0.5$. Full/empty: up/down. Fig. 4.



Pipe experiment (Poumaere *et al.*, 2014).



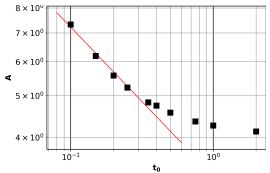
Up and down branches vs. t_0. Diamonds: creep steady states. Fig. 5.

With no structural variable:

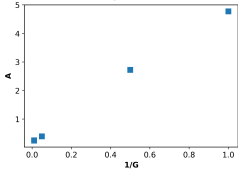
- Reversible only above yield
- Velocity plateaus below yield, decreasing with t_0
- A cusp: the flow reverses, at $\mathrm{Bn} \approx 1$ for all t_0

The hysteresis here is purely elastic

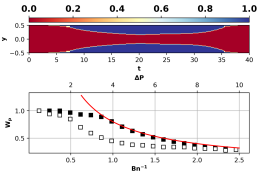
our work



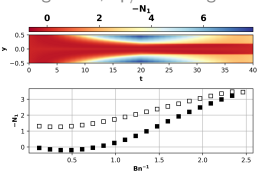
$A = 3.25\, t_0^{-0.34}$, Fig. 6.



A grows linearly with $1/G$, Fig. 9.



Plug width, up/down. Fig. 7.



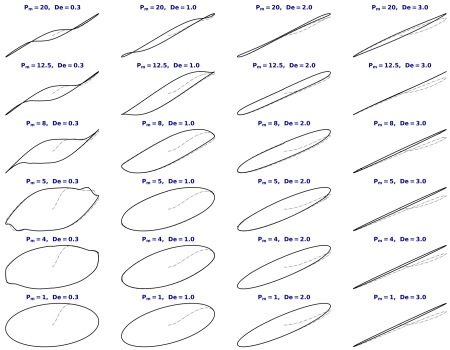
First normal stress difference. Fig. 8.

- Delayed re-solidification of the plug on unloading
- Elastic stresses persist near the walls: N_1 does not return to zero

In this model, elasticity is the *only* possible source of hysteresis.

Oscillations: the Pipkin space of the channel

our work



Lissajous–Bowditch figures (ΔP, U) vs. amplitude P_m and Deborah number De. Burghelea & Moyers-González (2025), Fig. 13.

- Small P_m: ellipse, elastic solid
- P_m just above P_c: partial yielding per cycle
- Large P_m: self-intersections, an elastic overshoot effect
- Rising De: more elongated shapes

Compare with the measured rheometric Pipkin space of Putz & Burghelea.

Saramito versus experiments: a fair scorecard

our work

Feature (Carbopol)	SRM-HB
Elastic solid below yield	✓
Plateau, decreasing with t_0	✓
Hysteresis; reversible in F	✓
Recoil cusp, flow reversal	✓
Normal stresses, tensorial form	✓
Power-law decay of A with t_0	✓ (form)
Value of the exponent α	differs (see right)
Thixotropy, structural ageing	absent
Wall slip	needs a separate law

Hysteresis exponent α

Rheometer (rough walls): 0.98–1.17
Pipe, with slip: ≈ 0.1
SRM-HB channel, no slip: 0.34

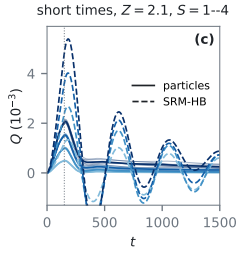
The paper points to slip and pipe 3D effects as possible reasons for the gap with the pipe.

Open: how to separate elastic from structural contributions to hysteresis?

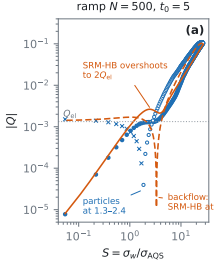
Back to the particles: a matched Saramito model

our work

Matching: ρ, H, G from the packing; τ_y, K, n from its local flow curve. Then nothing is fitted to unsteady data.



Loading from rest, $Z = 2.1$, $S = 1$–4. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 4c.



Ramp. Points: particles; lines: SRM-HB. Burghelea *et al.*, manuscript for Phys. Fluids (2026), Fig. 15a.

Shared: a peak at a load-independent time, $Q_{\max} \propto S$; backflow; the same lag loop.

Different:

- SRM-HB rings at period $\approx 2H/c_s$; the overdamped packing never does
- Overshoot: $1.6\,Q_{\mathrm{el}}$, against $0.62\,Q_{\mathrm{el}}$
- SRM-HB flows once $\sigma_w > \tau_y$, with no delay

What a better continuum model needs, according to the particles

our work

Test	Absent from SRM-HB
Creep	No ringing; start threshold and delays
Ramps	No loading overshoot; earlier backflow
Plug, N_1	Plug hysteresis in slow ramps; N_1 4–5× smaller; residual N_1 of random sign
Thresholds	Stop ≈ 0.3 below stability at all H
Oscillations	Transition at S_{start}, not τ_y; dissipation below yield

The **loop area** changes by under 5 % with any of these additions: a poor discriminator between models.

1. **Solid-state dissipation.** The plug behaves as a Kelvin–Voigt solid with $\eta_u \approx 21$, 200× the solvent viscosity that fits the flow curve, 14× the viscosity of the flowing channel.
2. **History-dependent relaxation.** One Maxwell mode in the solid fits loading ($G_1 + G$ agrees with the affine modulus to 6 %), but fails on unloading. Hence a structure variable, not more modes.
3. **A stop condition** tied to the flow state.

The circle closes: particles do not replace Φ; they show where it is needed and what it should describe.

LTeN

The whole landscape

	Oldroyd-B	Bingham	HB	Putz–Burghelea	de Souza M.–T.	Saramito
Shear thinning	–	–	✓	✓	✓	✓
Elasticity	✓	–	–	✓	✓	✓
Yield stress	–	✓	✓	✓	✓	✓
Normal stresses	✓	–	–	–	✓	✓
Hysteresis around yielding	–	–	–	✓	✓	✓
Structural kinetics	–	–	–	✓	✓	–
Tensorial	✓	✓	✓	–	✓	✓
Parameters	3	2	3	∼8	∼10	5

A rough guide to choosing

Slow, steady flows: HB. Unsteady flows, complex geometry: Saramito.
Rheometric tests near yield: Putz–Burghelea.
Thixotropy in complex flows: de Souza Mendes–Thompson.

A caution

A good fit of a flow curve does not validate a model. Test it on a different flow.

Take-home messages

1. Shear thinning, elasticity and yield stress are measurable, separate behaviours, but many real materials show them together.
2. Maxwell gives the concept of relaxation; Oldroyd-B turns it into a proper 3D model, via the upper-convected derivative.
3. Herschel–Bulkley is the steady limit. It describes $t_0 \to \infty$ well, and misses all the rest.
4. The transition is gradual and history dependent: S, S+F, F, with hysteresis $A \propto t_0^{-\alpha}$. Always state your protocol.
5. Structural models add a kinetic equation; Saramito adds elasticity with a yield criterion. Both reproduce hysteresis, by different mechanisms.
6. Particles with no structure at all already give an HB law, elastic branches, a lag loop and two thresholds. Add Φ where these fail: the flow history of the solid.
7. Keep the model as simple as possible, "but not too simple", and validate it on an independent flow.

Exercises

1. **Maxwell.** Insert $\gamma = \gamma_0 e^{i\omega t}$ into the Maxwell equation to derive G', G''. Where do they cross?
2. **Oldroyd-B.** Derive $N_1 = 2\eta_p \lambda \dot{\gamma}^2$ in steady shear. In uniaxial extension, show that τ_{zz} diverges at $\lambda\dot{\epsilon} = 1/2$.
3. **Herschel–Bulkley.** Derive the channel profile, the flow rate for Bingham, and the critical $\Delta P/L$.
4. **Elastic plateau.** For $\tau = Bt < \tau_y$, show that structural and Saramito models both give $\dot{\gamma} = B/G$. How do you then measure G from a ramp?
5. **Putz–Burghelea.** Find the fixed points of the kinetic equation for $\delta = 0$ and their stability. Then include a small $\delta > 0$.
6. **Saramito.** Recover the Oldroyd-B limit. Derive τ^* for yielding from rest in simple shear, and the plug width in a channel.
7. **Particles.** Derive $\bar{u} = \sigma_w H/6G$ for a clamped elastic slab. For the lag law, show that $A \propto T^{-1}$ in slow ramps. Run `code/particle_sketches.py`.
8. **Numerics.** Run `code/model_figures.py`. Change t_0 and w. What sets the area of the loop?

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